Python NETGEN .VOL Converter Short Guide

Read Original Article by Dip28 here:
http://dip28p.web.fc2.com/calculix/netgen2calculix/index.html

(or in PDF copy in the folder: \c4w\Hlp\conv\Netgen_by_Dip28.pdf).

Short Description (for bodes):

1. Run Netgen. Go to File>Load Geometry and load geometry. IGES file may require some correction. Use the menu: Geometry>IGES/STEP Topology Explorer/Doctor for that.
2. Go to menu: Mesh > Generate Mesh. Or set up options first at Mesh>Meshing Options (for example use mesh-size grading at mesh size tab)
3. After meshing, mark faces by changing their numbers (not for all, just to apply BC then). Menu: Mesh>Edit boundary conditions... Then double click on face and change the number (bc property). For example set up 21111 instead of 2, etc.
4. If you have rough mesh you can use: Refinement > Refine Uniform or other options.
5. Refinement > Second Order **(it's a must!!! current python script doesn't support linear elements. Program will be crashed without this option.**
6. File > Save Mesh
7. You can also use export features. First, select the filetype: File > Export FileType >>... Then File > Export mesh (write the name with extension). Abaqus format is supported (just clean mesh without groups), also GMSH v2 format is available.

See the example below:

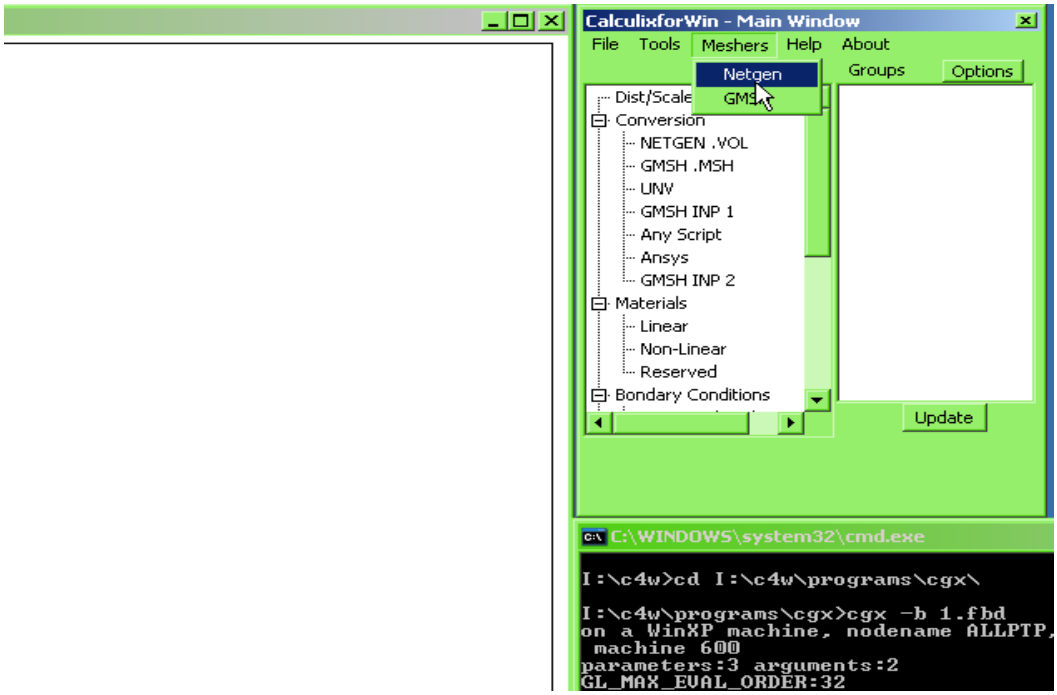


Fig. 1 Running NETGEN

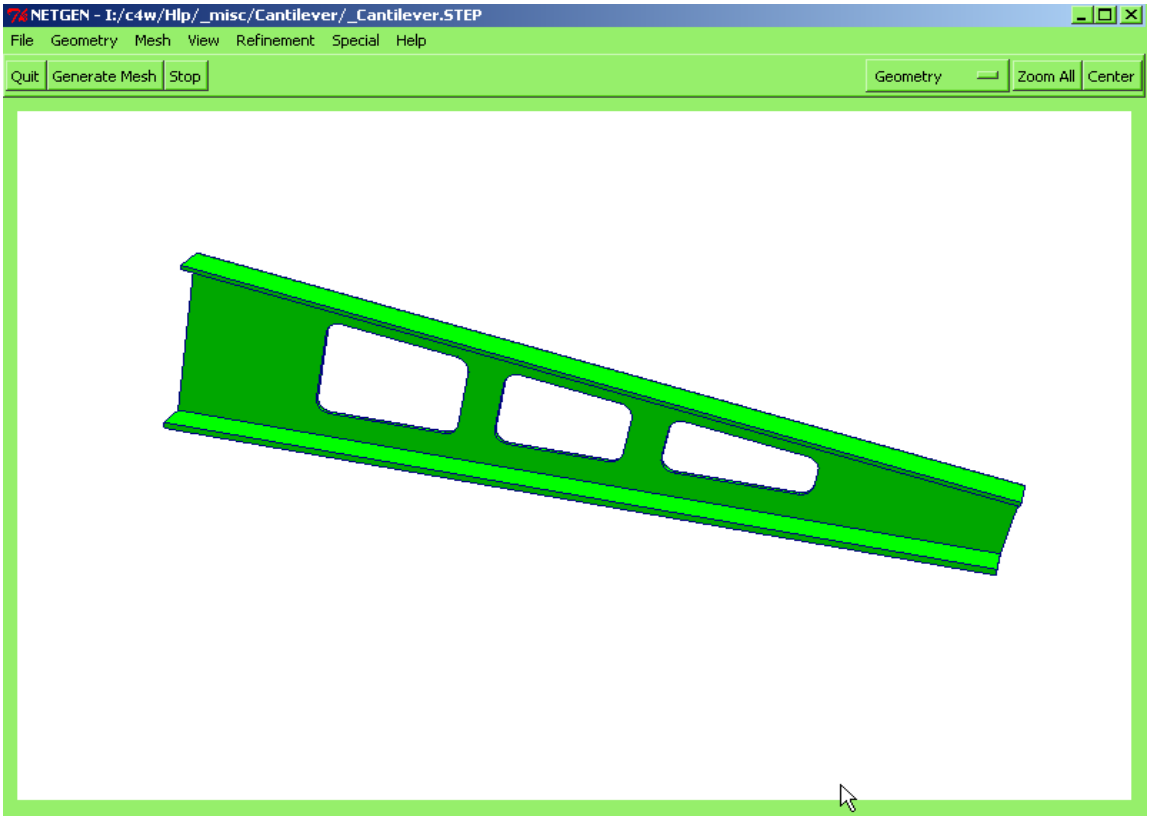


Fig. 2 - Load STEP geometry

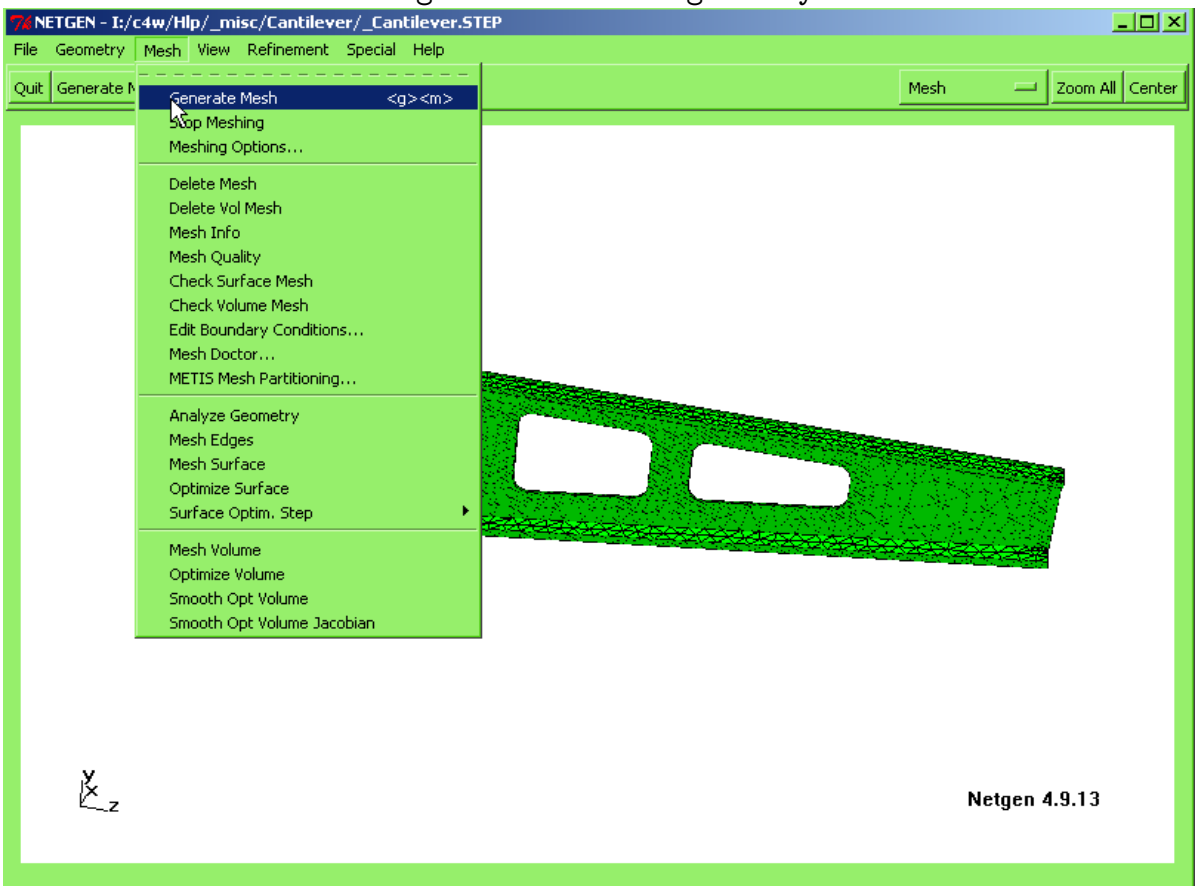


Fig. 3 - Generate mesh

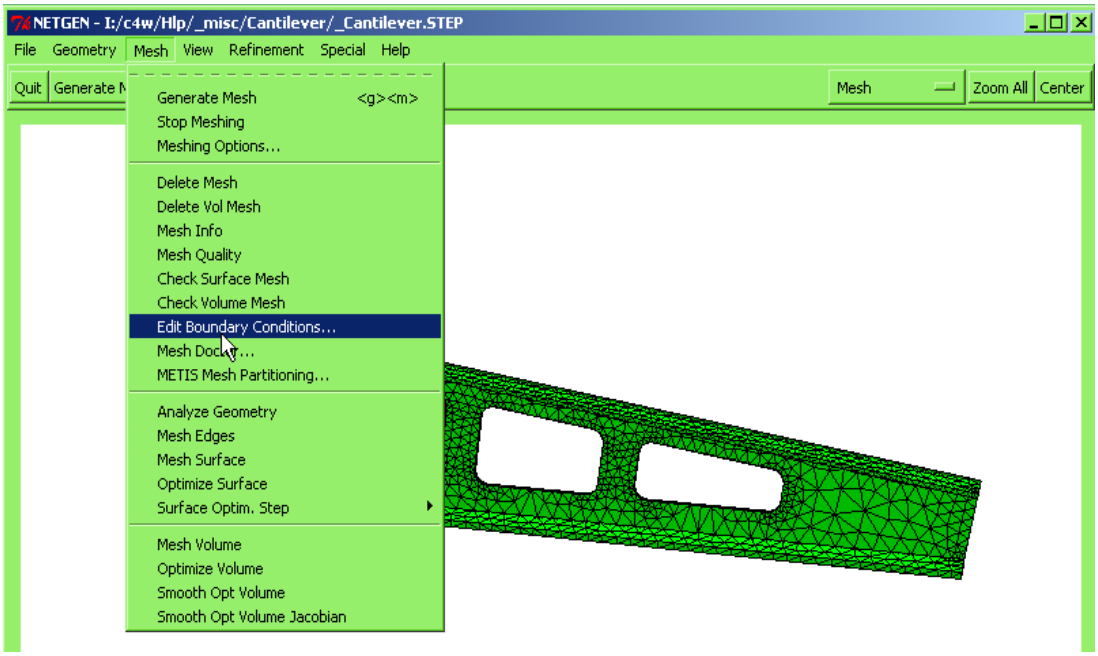


Fig. 4 Edit boundary conditions

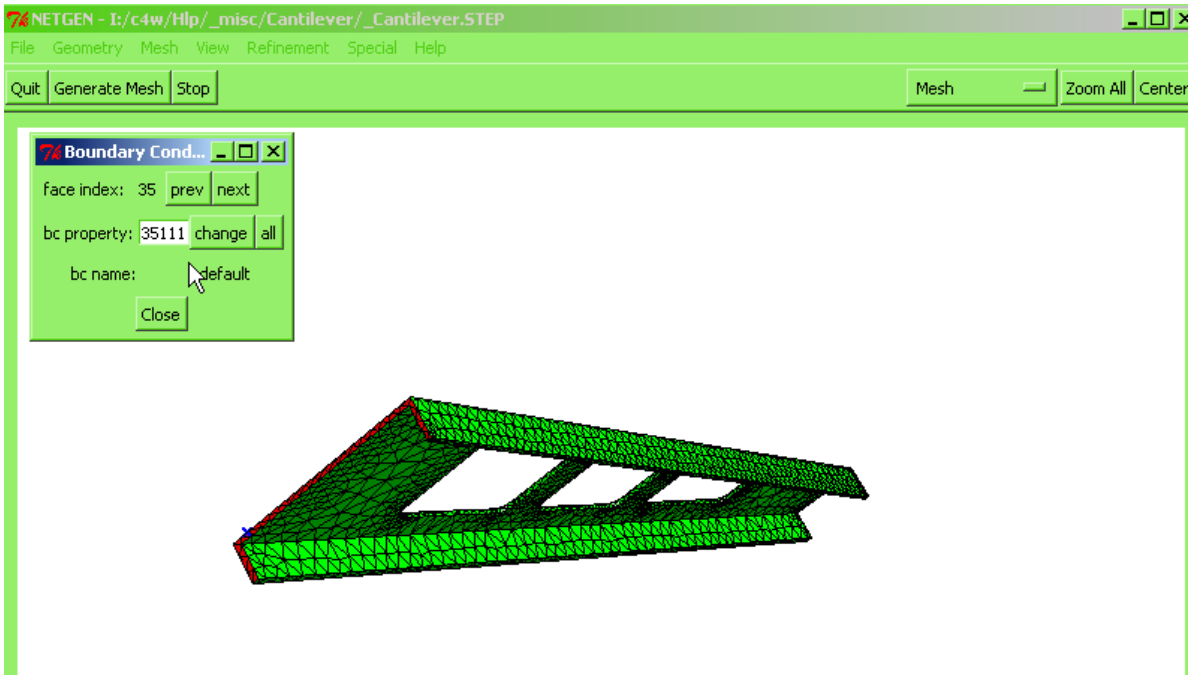


Fig. 5 - Double Click on face and change the number to mark it for future work (this face for fixed support)

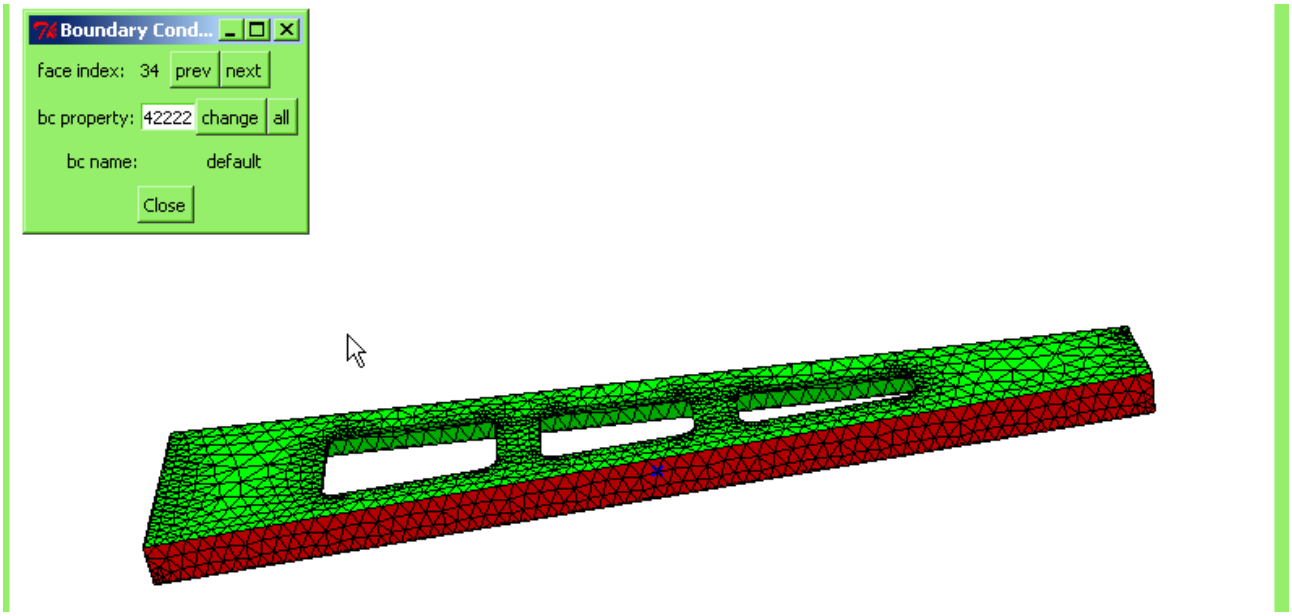


Fig. 6 - Mark face for load

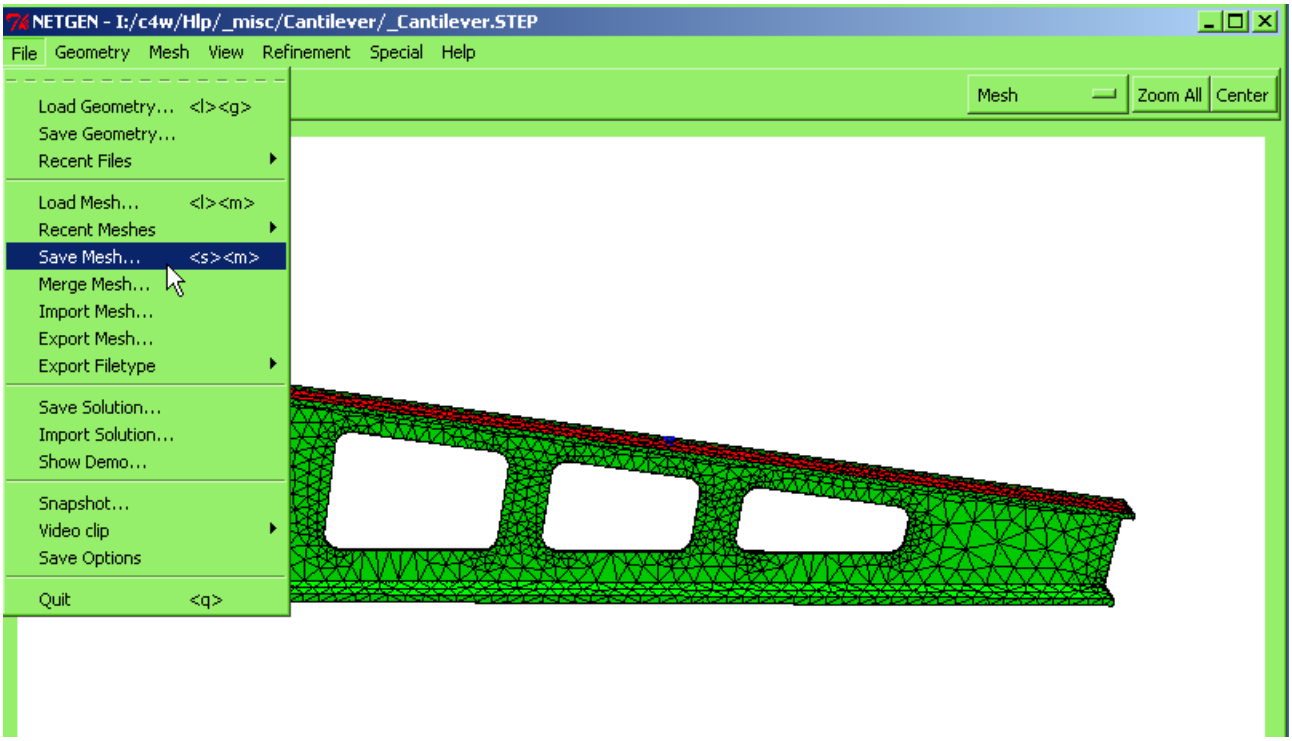


Fig. 7 Second order is necessarily

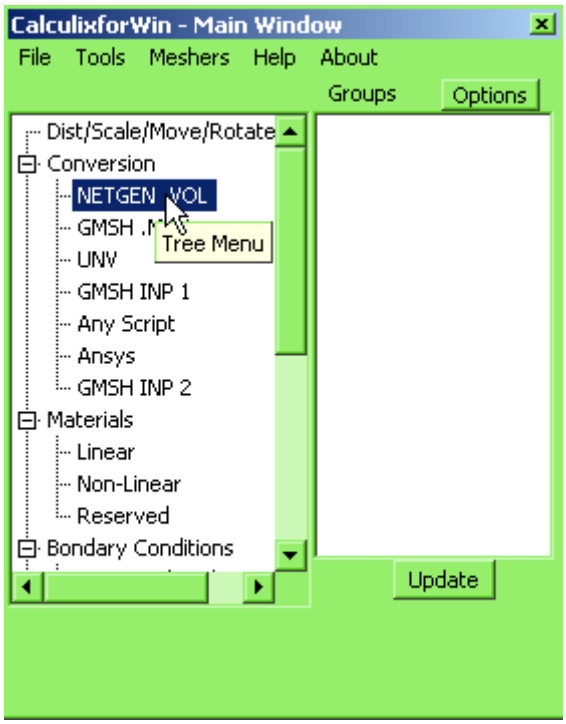


Fig. 8 - In CalculiXForWin - open converter window

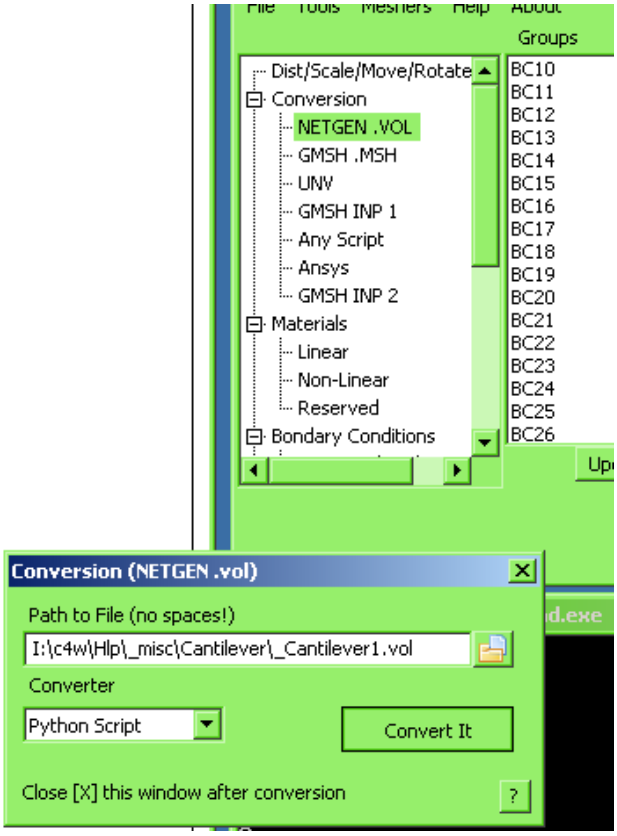


Fig. 9 Set path to .VOL file and press "Convert It" After conversion close window manually

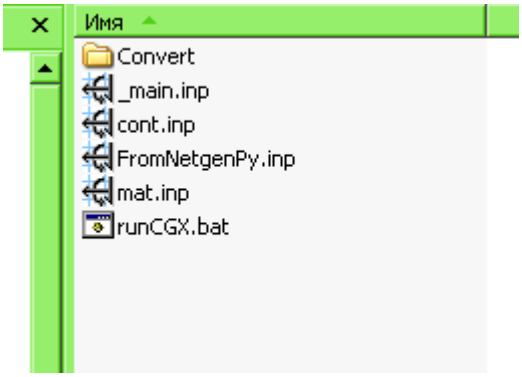


Fig. 10 - The converted INP file will be stored in the working folder

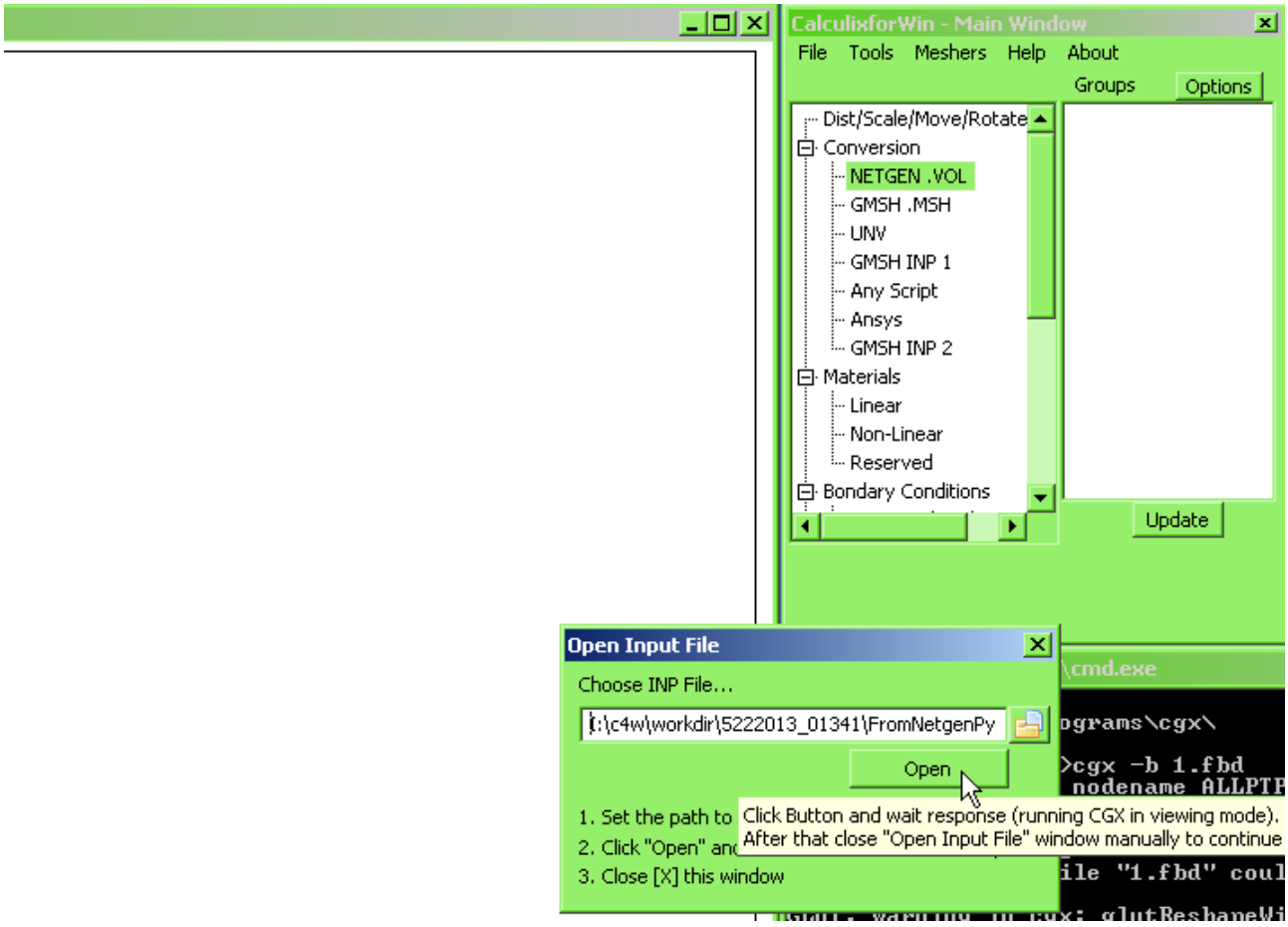


Fig. 11 - Go to the File> Open INP File and set path to the "FromNetgenPy.inp" in the working folder

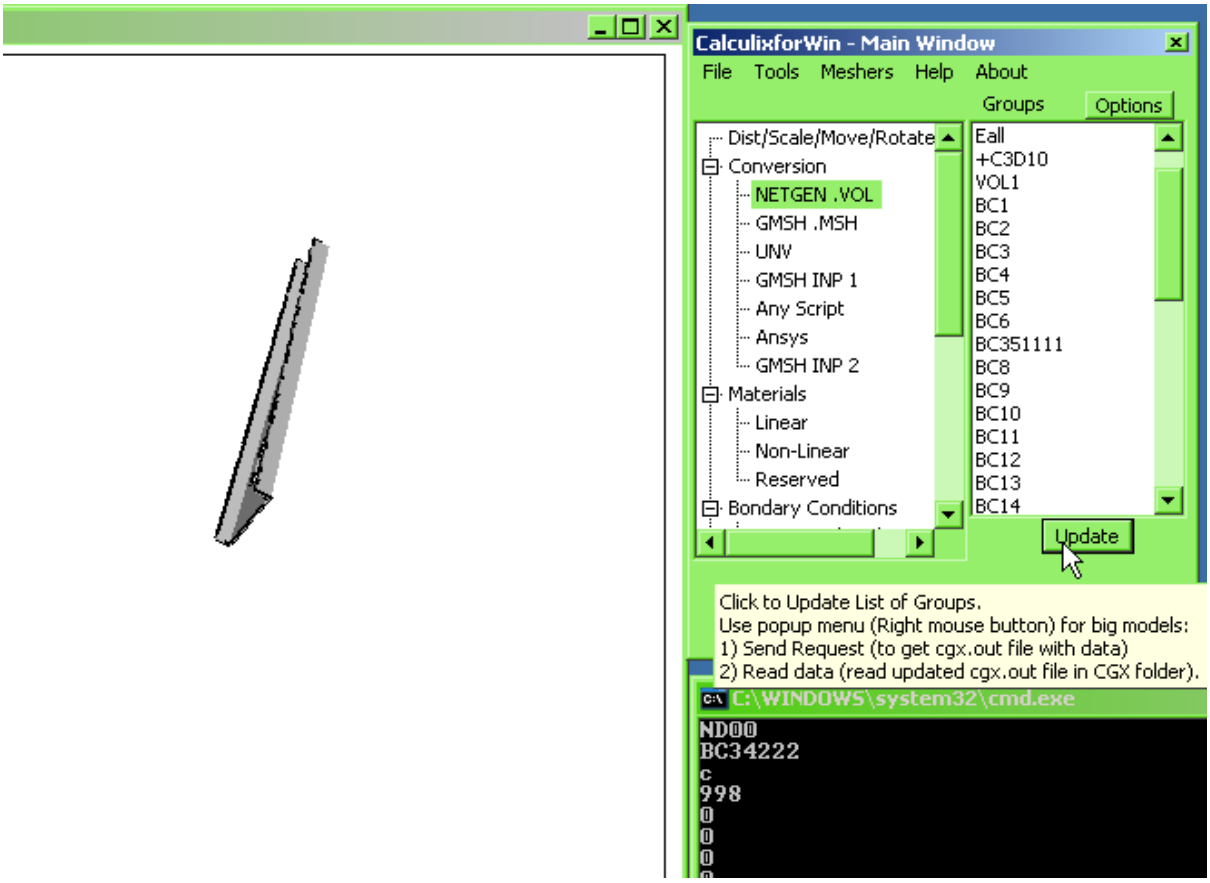


Fig. 12 Update the groups after opening

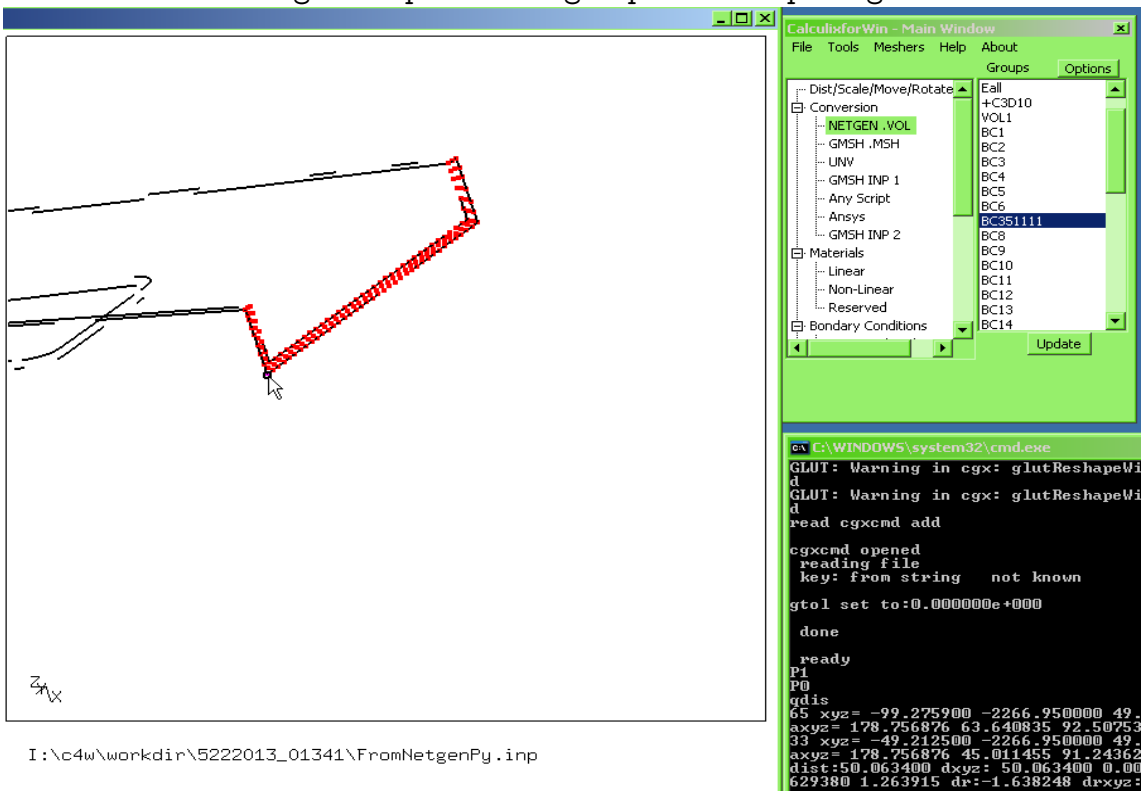


Fig. 13 Check the distance with using "qdis" CGX card. We have STEP file with inches, but now the dimension is "50" instead of "2". The NETGEN had scaled STEP geometry into mm. We have to scale it back to use Imperial units.

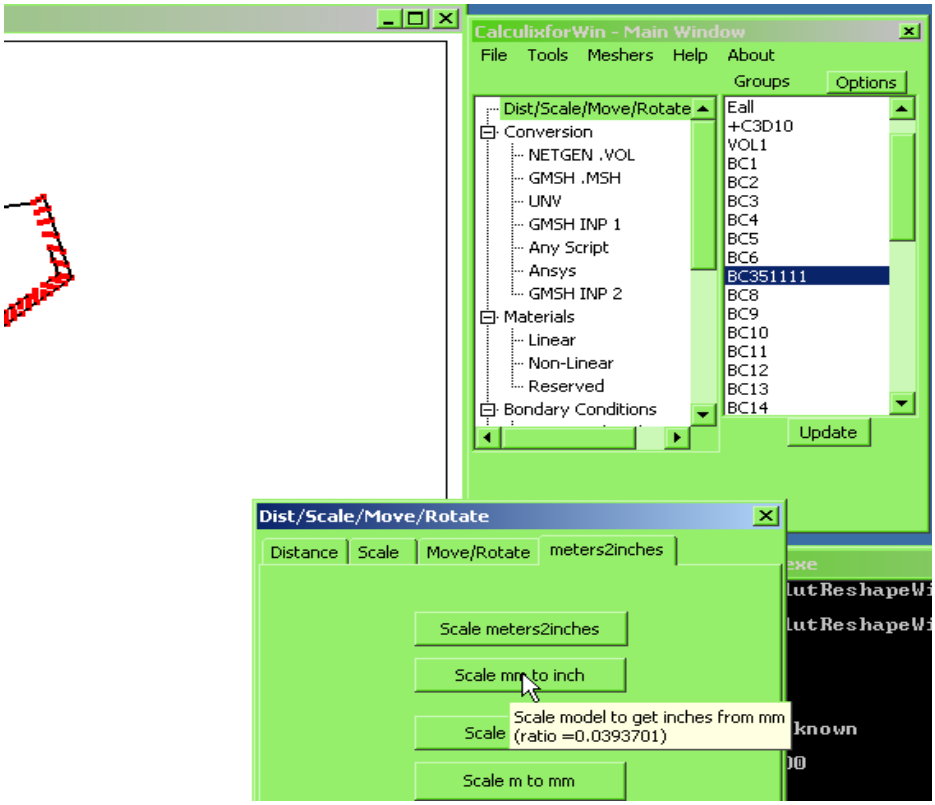


Fig. 14 Open "Dist/Scale/Move/Rotate" window and scale from "mm" to "inch"

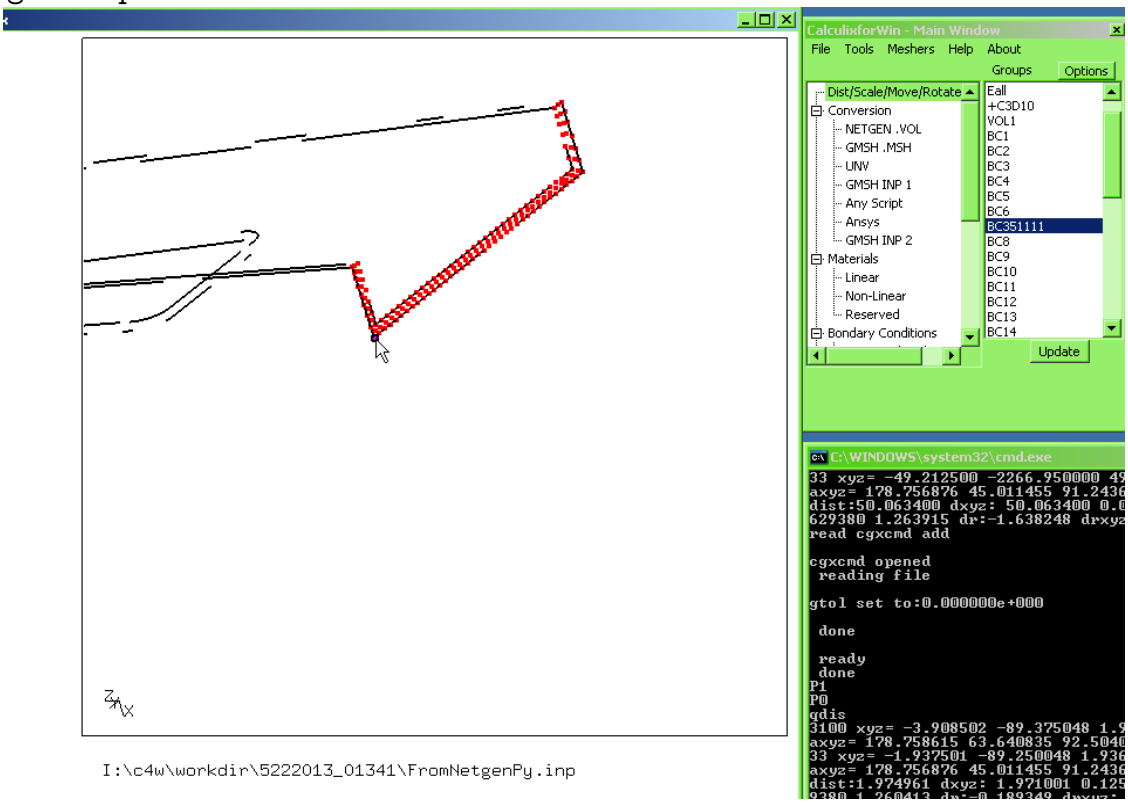


Fig. 15 Now we have "dist=1.97" what is correct (use "qdis card in CGX")

NOTE: DON"T FORGET TO UPDATE THE GROUPS TO CHANGE DATABASE PROPERTIES IN CALCULIXFORWIN!!

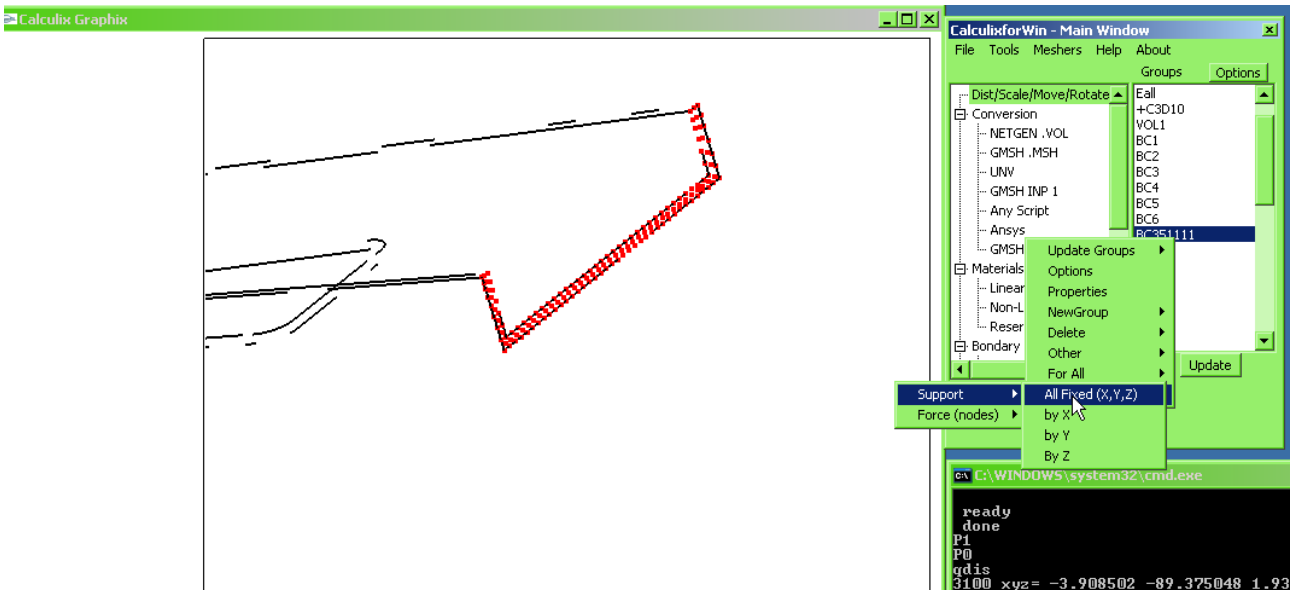
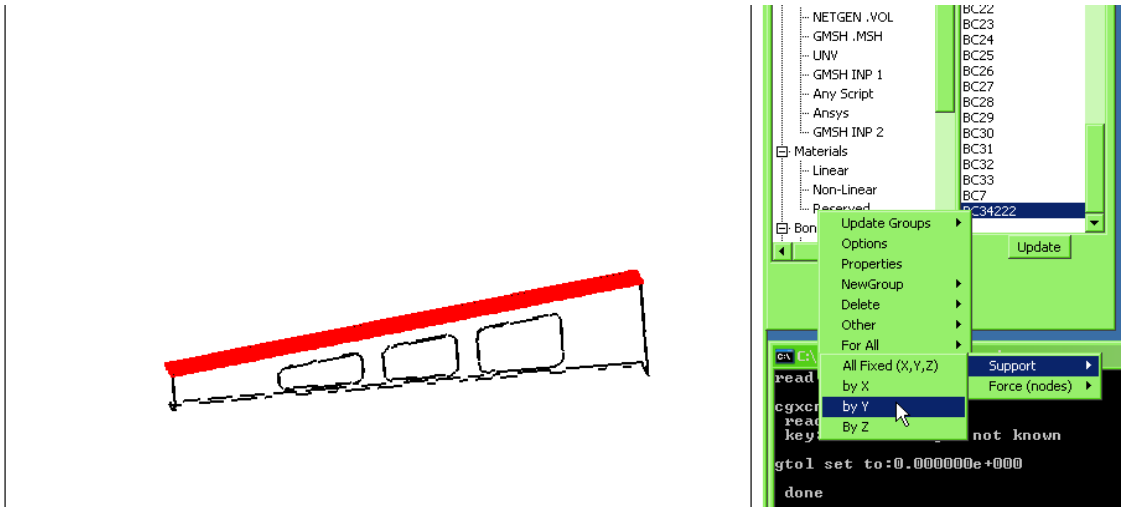


Fig. 16 Apply fixed support to the selected group



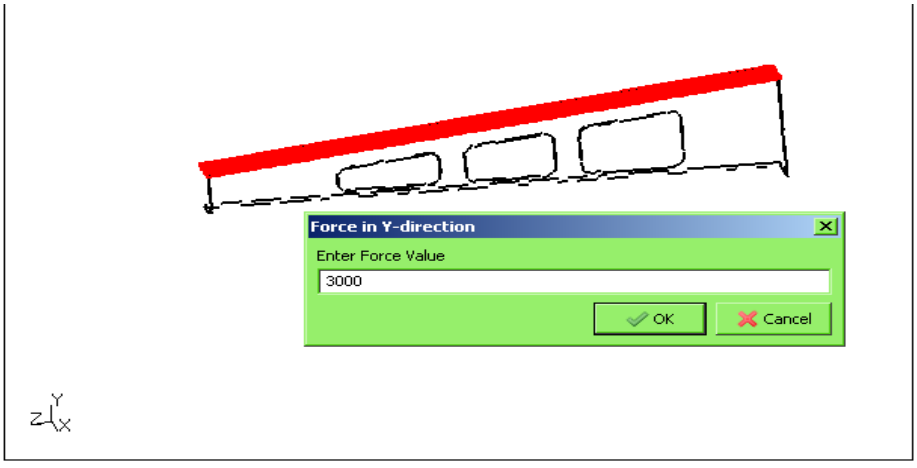


Fig. 17 Apply the load to the nodes of group selected

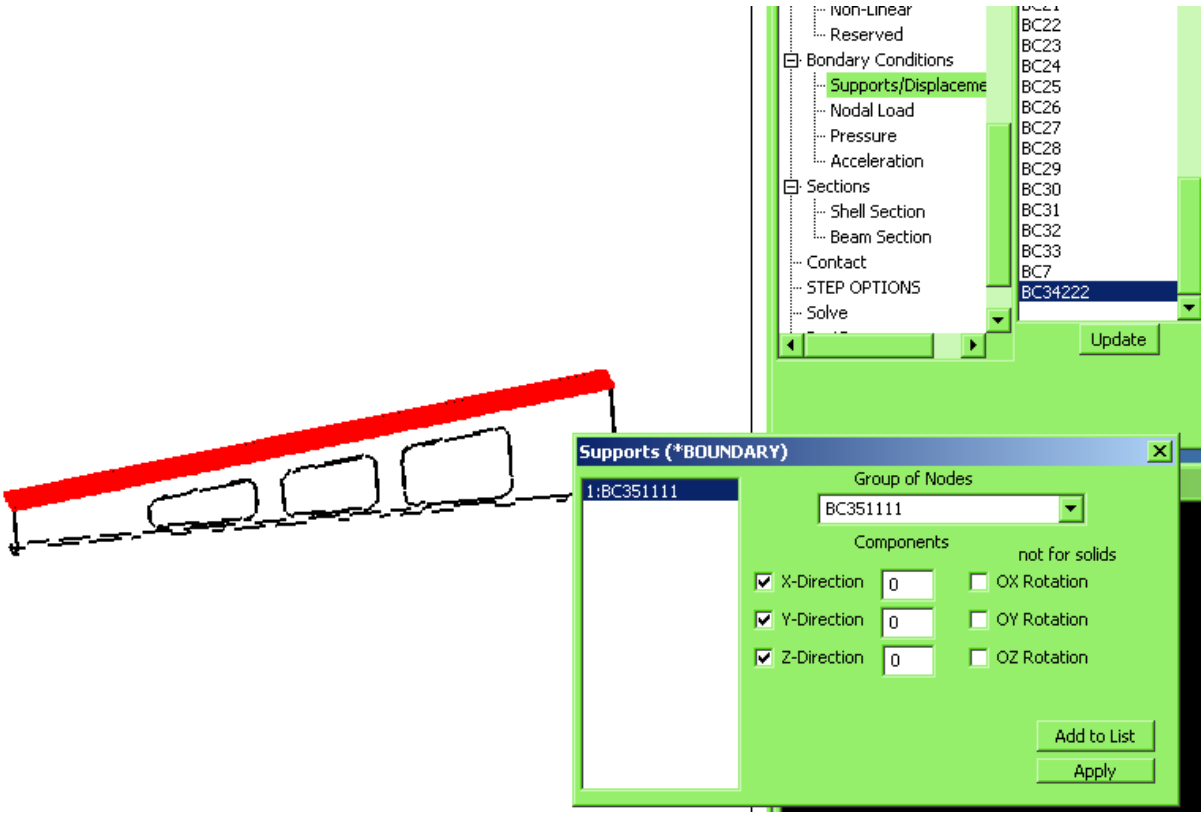


Fig. 18 Check BC in the database

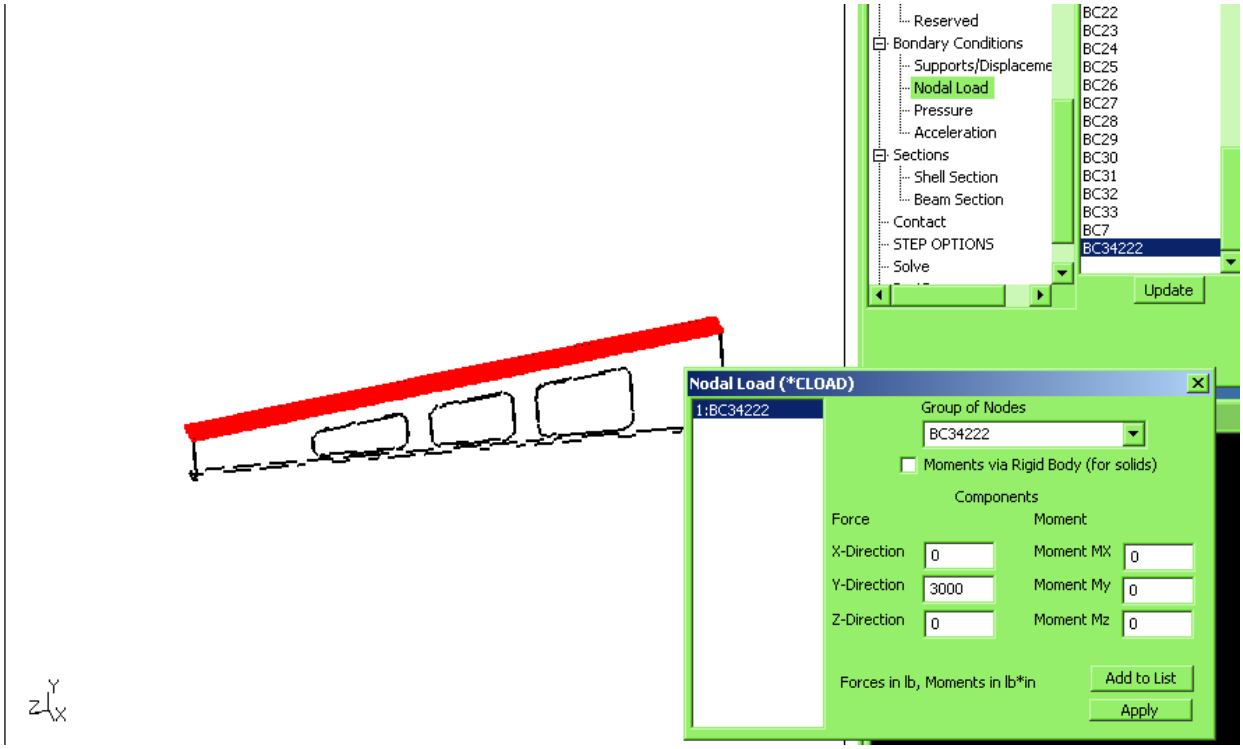


Fig. 19 Check the load in the database

You can also check the inp files updated in the working folder

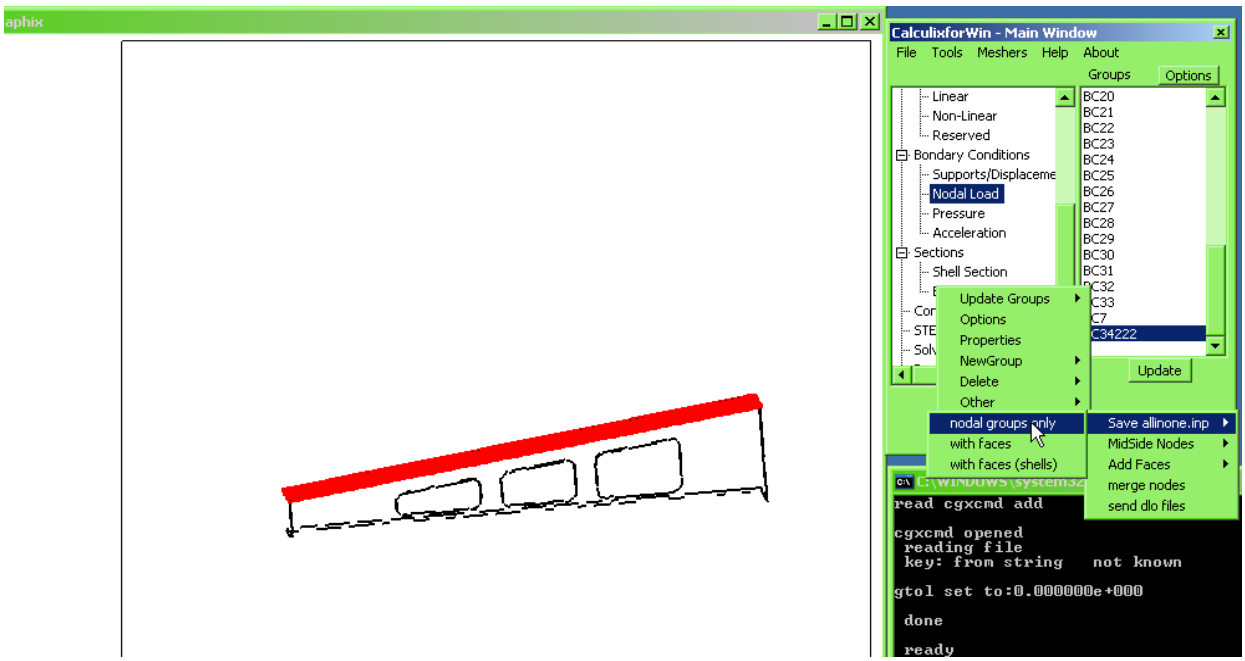


Fig. 20 Save updated (scaled) mesh in the allinone.inp file (here we can use "nodal groups only" option)

This procedure can take a **long time**. Watch for GGX Output window to be sure that the process of saving is done/ready to continue

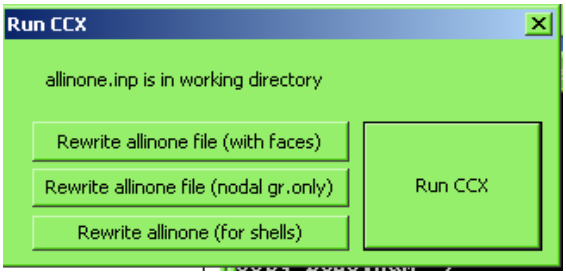


Fig. 21 - Go to the "solve" in tree menu and run CCX (CalculiX solver)

It will run main.inp file in the working folder. The solution will be stored in the "main.frd" file

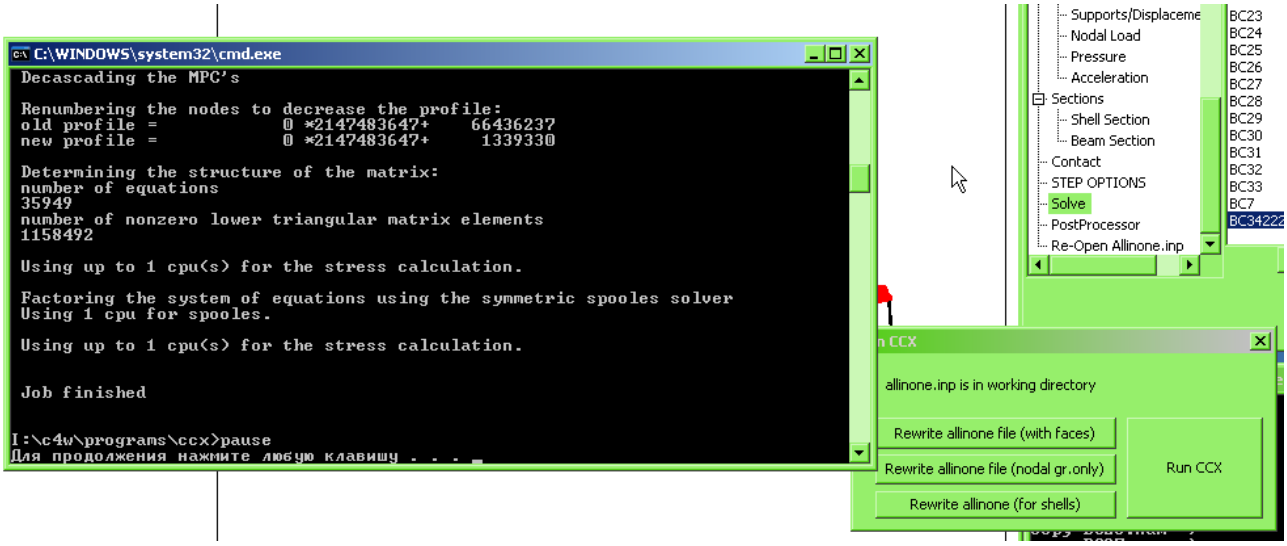


Fig. 22 - Job finished. Go to the working folder and check inp files with data if something wrong.

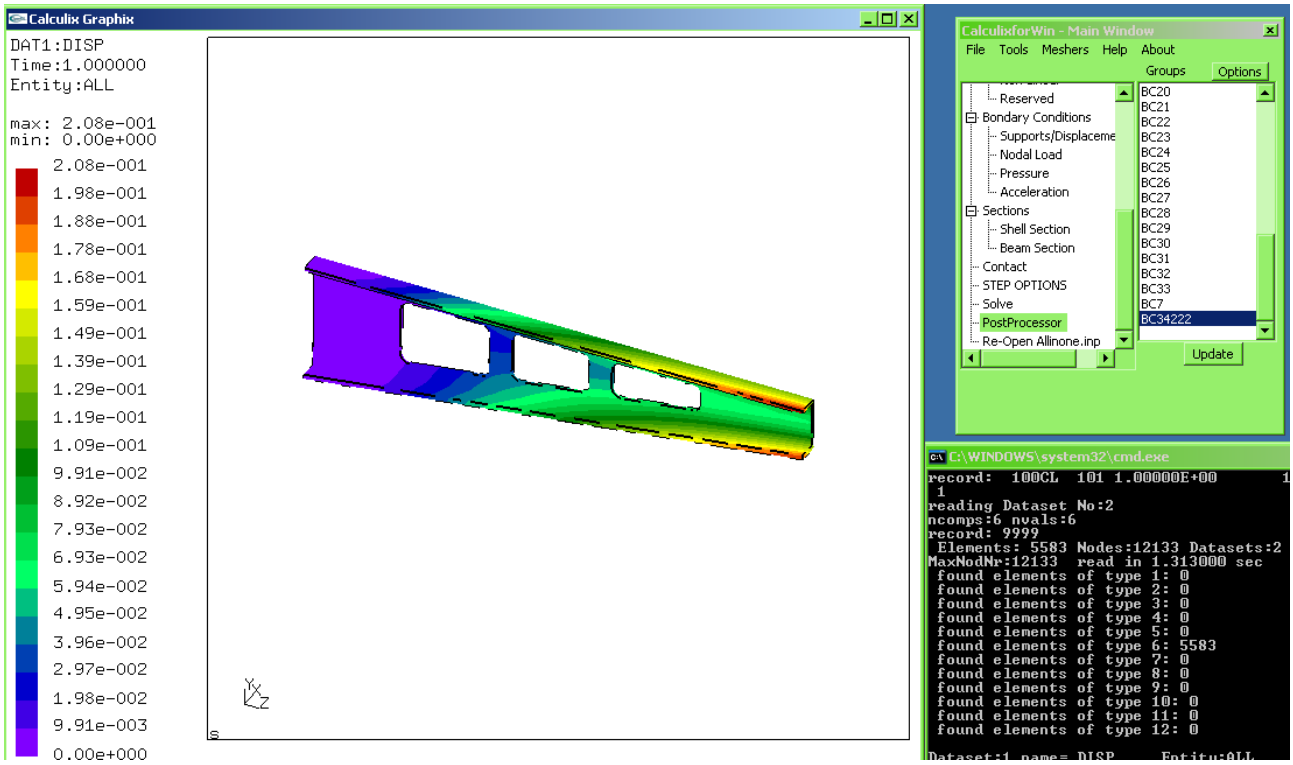


Fig. 23 - Run CGX in Post-processing mode (it will run _main.frd in the working folder). Here are displacements in inches

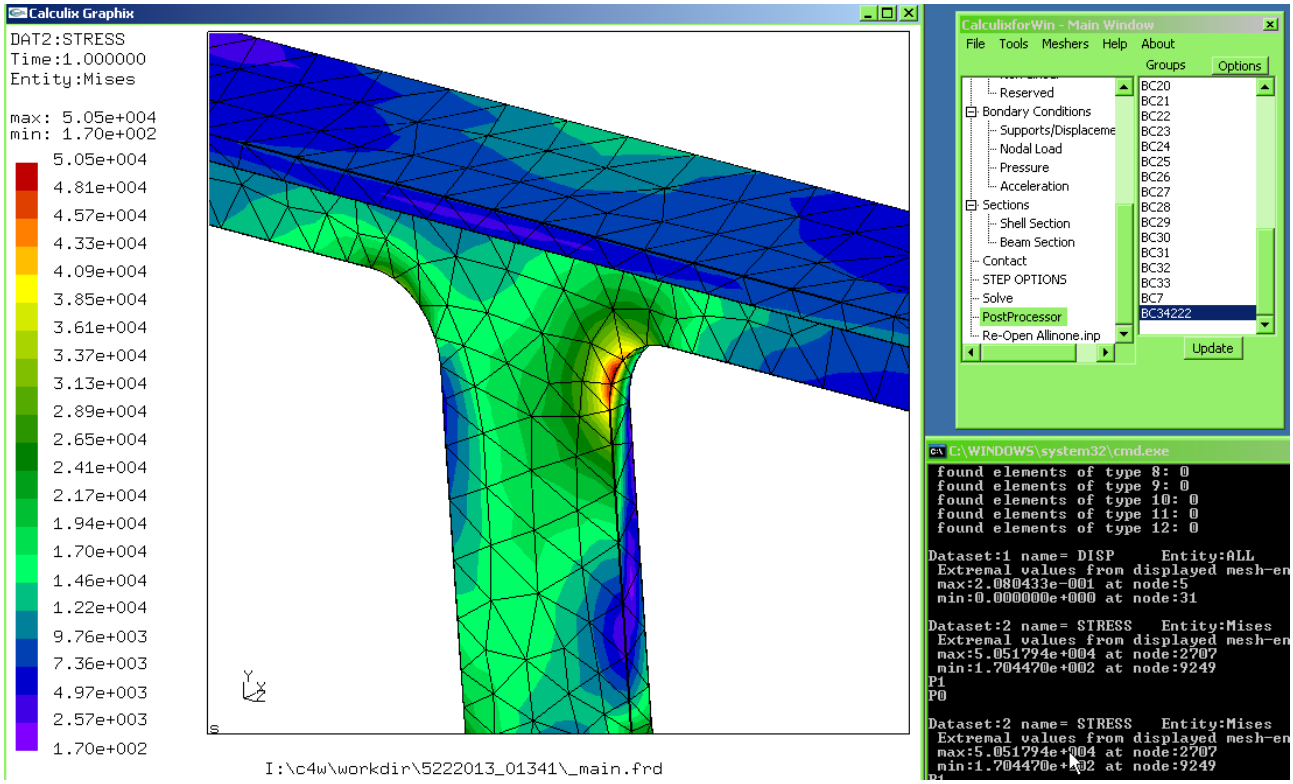


Fig. 24 - Max. Von Mises stress are localized